



Operations Research

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Exercise Sheet 10

Training Exercises

Exercise T10.1. *Gradient Descent vs. Newton's Method*

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as $f(x_1, x_2) = x_1^2 + x_1x_2 + 1.5x_2^2 + x_1 - 4x_2 + 3$.

- (a) Compute the first iteration $x^{(1)}$ of the Gradient Descent method starting from $x^{(0)} = (1, 1)^T$ with a constant step size $\eta = 0.5$.
- (b) Compute the first iteration $x^{(1)}$ of Newton's Method starting from the same initial point $x^{(0)} = (1, 1)^T$.
- (c) Discuss the behavior of Newton's Method on this function compared to Gradient Descent.

Exercise T10.2. *Lagrange multipliers*

Consider the following function and the corresponding constraint:

$$f(x, y) = -e^{x^2+y^2} \tag{1}$$

$$s.d. \quad g(x, y) = 2x + y = 5 \tag{2}$$

- (a) Show that f is concave.
- (b) Set up the corresponding Lagrange function $L(x, y, \lambda)$ for the restricted problem.
- (c) Determine the critical points (x^*, y^*, λ^*) of the Lagrange function. Decide and explain whether (x^*, y^*) is a saddle, maximum, or minimum point of the function f .
- (d) Let (x', y', λ') be a point such that

$$\left(\frac{\partial L}{\partial \lambda} \right) (x', y', \lambda') = 0.$$

What can we then say about $g(x', y')$?

Exercise T10.3. *Non-linear optimization - KKT conditions*

Consider the following non-linear program.

Use without proof that a regularity conditions holds, i.e., the KKT conditions are necessary.

- (P) $\max \quad \sin(x_1) + \cos(x_2)$
 $\quad \quad x_1 + x_2 \leq 2$
 $\quad \quad 2x_1 - x_2 \geq 1$
 $\quad \quad x_1 \geq 0$
 $\quad \quad x_2 \geq 0$
- a) State the KKT conditions for (P). How do the KKT conditions change if we minimize instead of maximize?
- b) Using a), show that the point $x = (1, 1)$ is not a local maximizer for P.
- c) Consider the multiplier $\mu = 0$. Using a), derive the corresponding point x and check whether x is a local maximizer for (P). If so, is x also a global optimum and is it unique?
- d) Bonus (hard): Show that $x = (1, 1)$ is a local minimizer.